CY4001: Industrial Chemistry

Unit 4:Polymer Chemistry
Lecture 3 – Polymerization

Alan MacDiarmid

Alan MacDiarmid is often called the "father of conductive polymers" or "synthetic metals"

Incredible Journey from a Janitor to a Nobel Laureate

He took up a part-time job as a "lab boy" or janitor at Victoria University of Wellington during his studies for a BSc degree

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He was then appointed demonstrator in the undergraduate laboratories.

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After completing an MSc in chemistry from the same university, he worked as an assistant in its chemistry department

↓

He won a Fulbright Fellowship to the University of Wisconsin–Madison. For PhD

↓

Professor in chemistry at the University of Pennsylvania, USA

↓

Nobel Laureate

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Polymerization

Polymerization is a process in which a large number of small molecules combine to form a bigger molecule with or without elimination of the small molecules.

Degree of Polymerization (DP)

- The size of a polymer molecule is represented in terms of degree of polymerization. The number of repeating units or monomers present in a polymer chain is known as degree of polymerization (n).

$$\textbf{Degree of Polymerization (DP)} = \frac{\textbf{Molecular weight of the polymeric network}}{\textbf{Molecular weight of the repeating unit}}$$

Example

$$4\,CH_2{=}CH_2 \longrightarrow -CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-$$

DP = 4

- **Oligo Polymers:** They are polymers with a low degree of polymerization and have a 500–5000 molecular mass range.

- **High Polymers:** They are polymers with high degree of polymerization and have a 10,000–2,00,000 molecular mass range.

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Types of Polymerization

Polymerization

- Addition Polymerization or Chain growth Polymerization
- Condensation Polymerization or Stepwise Polymerization
- Co-Polymerization
- Coordination Polymerization

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Addition or Chain-growth Polymerization:

It is a reaction that yields a polymer which is an exact multiple of the original monomeric molecule. Molecular weight of the polymer is an exact multiple of the monomer.

(i) Polyethylene is produced from ethylene.

$$n\,CH_2{=}CH_2 \xrightarrow[\text{Catalyst}]{\text{Heat/Pressure}} -(CH_2-CH_2)_n-$$

Ethylene — Polyethylene (PE)

(ii) Polystyrene is produced from styrene

$$n\,CH_2{=}\underset{\underset{C_6H_5}{|}}{CH} \xrightarrow{\text{Polymerization}} -(CH_2-\underset{\underset{C_6H_5}{|}}{CH})_n-$$

Styrene — Polystyrene

- Addition polymerization occurs in monomers that contain at least one double bond or triple bond.
- This type of polymerization takes place under suitable conditions, i.e. in the presence of a suitable catalyst and by applying appropriate pressure and temperature.

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Free radicals

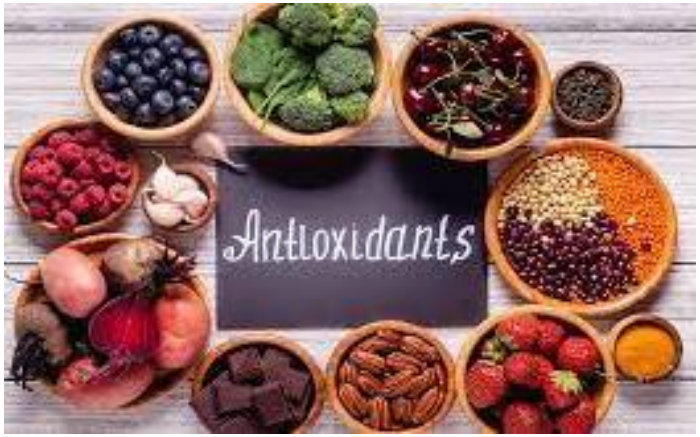
- A free radical is a molecular species which can contain one or more unpaired electrons in its atomic orbital and can exist independently.

- Free radicals are molecules that can be highly reactive and cause damage to other molecules, such as DNA, lipids, and proteins. They are often linked to multiple illnesses, including diabetes, heart disease, cancer and aging.

- Causes: Exposure to toxic chemicals (pesticides), air pollution, smoking, alcohol and fried foods

Anti-oxidants

Antioxidants are chemicals that interact with and neutralize free radicals, thus preventing them from causing damage. Antioxidants are also known as "free radical scavengers".



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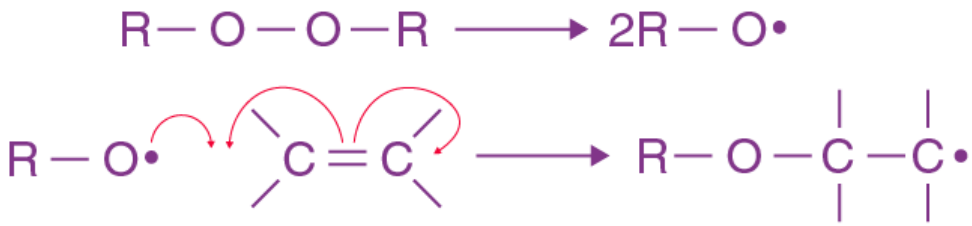
Mechanism of Addition Polymerization

Free-radical mechanism

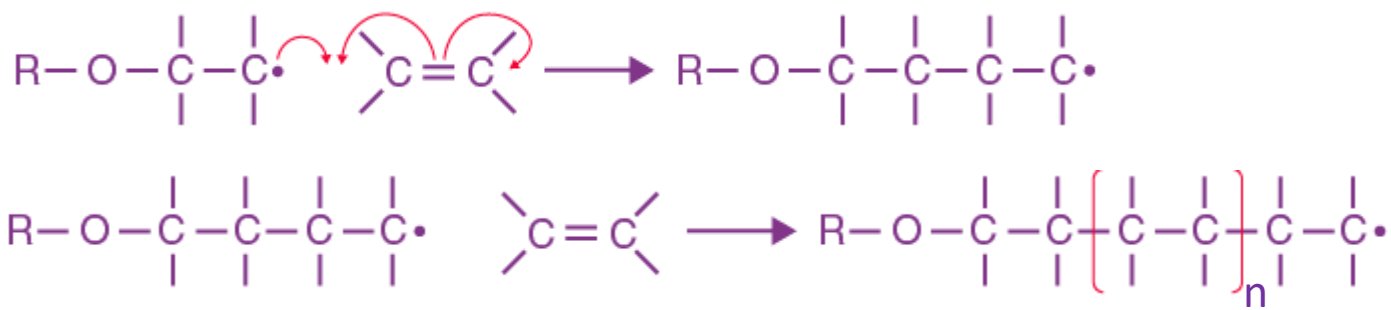
1. Chain Initiation (Free Radical chain)

One or two radicals are created from the initiating molecules by **homolytic dissociation**



2. Chain Propagation (expanding)

The formed radical reacts with another monomer to expand its chain length (propagating).



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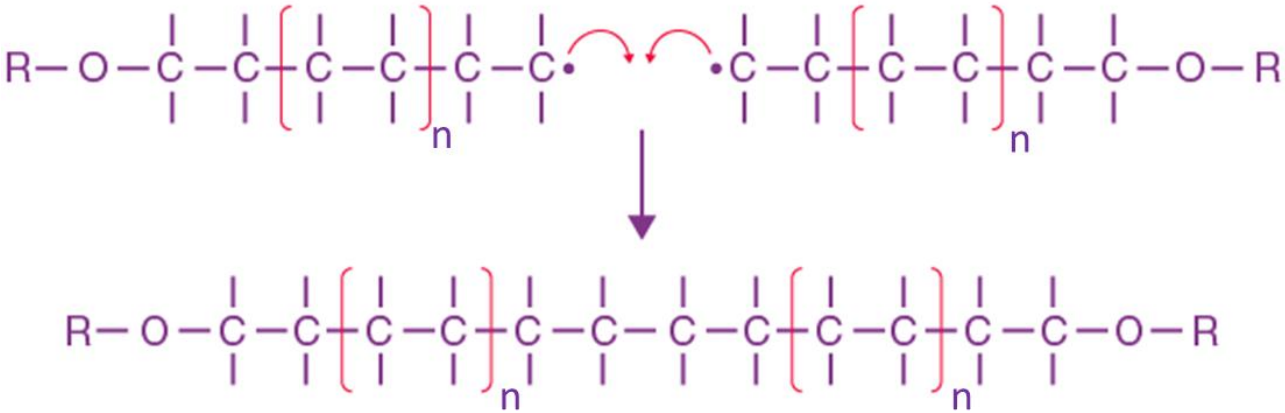
Mechanism of Addition Polymerization

Free-radical mechanism

3. Chain Termination

The growth of polymer chain is terminated either by a coupling reaction or disproportionation

Coupling reaction



Disproportionation reaction

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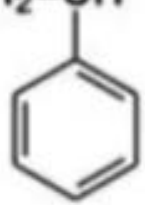
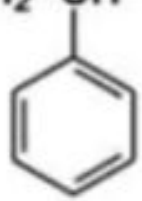
SOME COMMON ADDITION POLYMERS

example	monomer	polymer	uses
polyethylene	$CH_2{=}CH_2$	$-CH_2-CH_2-$	most common polymer bags, wire insulation, squeeze bottles
polypropylene	$CH_2{=}CH(CH_3)$	$-CH_2-CH(CH_3)-$	fibers, bottles, indoor-outdoor carpet
polystyrene	$CH_2{=}CH$ 	$-CH_2-CH-$ 	styrofoam, inexpensive molded objects: household items, toys
polyvinyl chloride (PVC)	$CH_2{=}CH(Cl)$	$-CH_2-CH(Cl)-$	synthetic leather, clear bottles, floor coverings, water pipe
Teflon	$CF_2{=}CF_2$	$-CF_2-CF_2-$	non-stick surfaces, chemically resistant items

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Condensation Polymerization or Stepwise Polymerization

This reaction involves the condensation of two functional groups present in the same kind of monomer or different kind of monomers by the elimination of small molecules like H_2O, HCl, NH_3, CH_3OH.

(a) Condensation Reaction Between two Different Monomers

$$n\,H_2N{-}(CH_2)_6{-}NH_2 + n\,HOOC{-}(CH_2)_4{-}COOH \xrightarrow{-2n\,H_2O} {-}(NH{-}(CH_2)_6{-}NH{-}CO{-}(CH_2)_4{-}CO)_n{-}$$

Hexamethylenediamine Adipic acid

Nylon 6 : 6 (Polyamide)

- The monomers should have two reactive functional groups.
- The polymerization proceeds by stepwise reaction between reactive functional groups.
- In some cases, both reactive functional groups could be present on the same monomer. In that case, it can lead to a self-polycondensation reaction.

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Condensation Polymerization or Stepwise Polymerization

(b) Self-condensation of a Monomer Having Different Functional Groups

- Self-condensation of ω-amino caproic acid

$$n\ H_2N\text{-}(CH_2)_5\text{-}COOH \longrightarrow \text{-}(HN\text{-}(CH_2)_5\text{-}CO)_n\text{-}$$

ω-amino caproic acid — Nylon-6

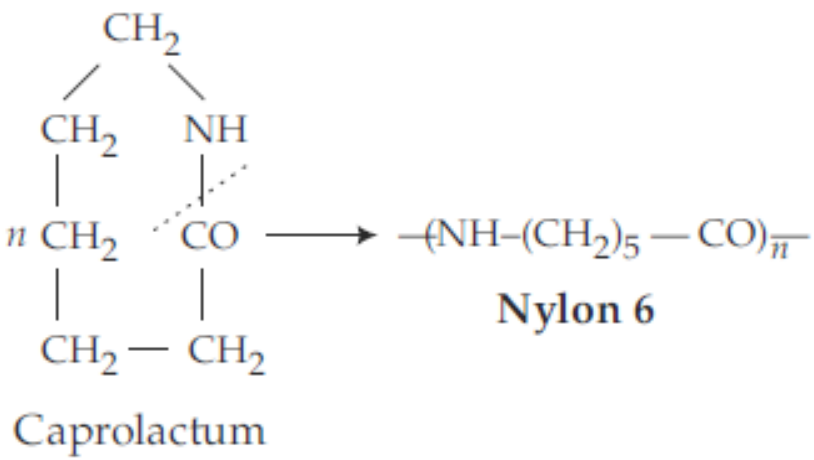
- Self-condensation of ω-amino decanoic acid

$$n\ H_2N\text{-}(CH_2)_{10}\text{-}COOH \longrightarrow \text{-}(HN\text{-}(CH_2)_{10}\text{-}CO)_n\text{-}$$

Nylon-11

(c) Ring-opening Polymerization

In some cases, condensation polymers are obtained without the elimination of small molecules but it takes place by just opening of cyclic compounds. This type of polymerization is known as ring-opening polymerization.



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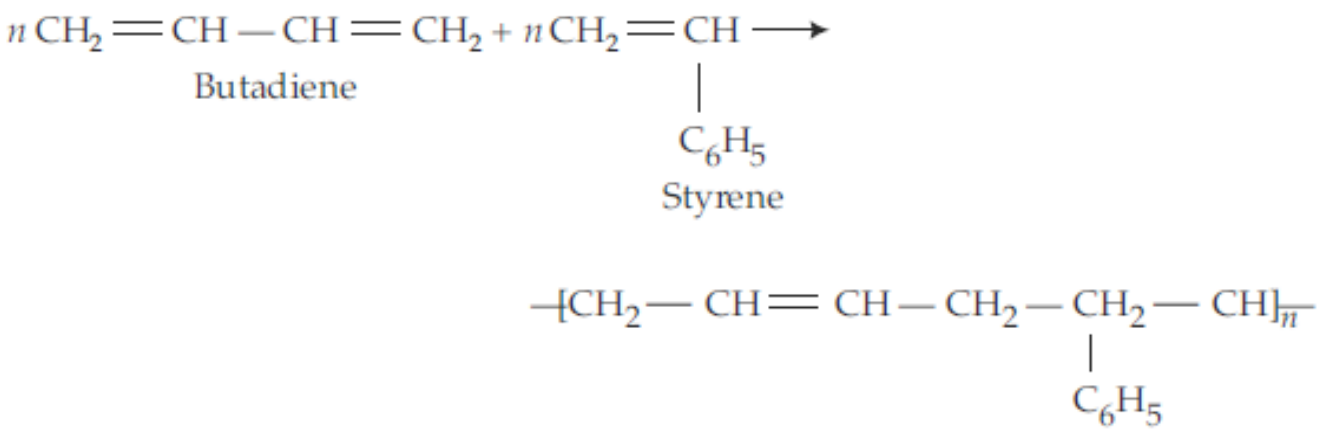
Co-polymerization:

- Co-polymerization is the joint polymerization of two or more different monomeric species. High-molecular-weight polymers obtained by copolymerization are called co-polymers. In this process, no side products are formed.

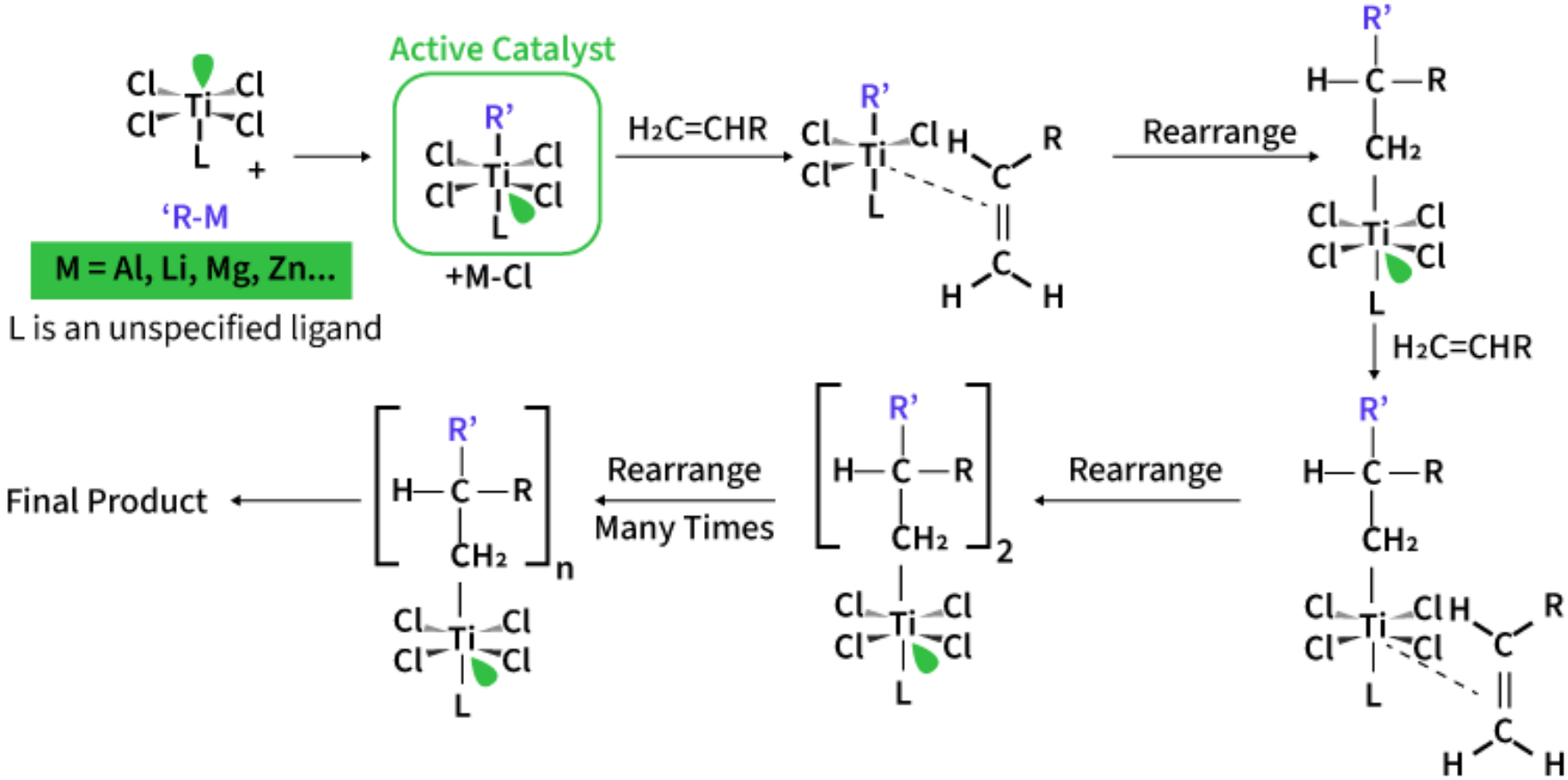


Styrene Butadiene Rubber (SBR)

- Chemical and geometrical shapes of polymers can be varied by this reaction and hence it is used to produce unsymmetrical or irregular shape polymers.

- Therefore, co-polymerization is superior than other polymerization reactions because it can control various polymeric properties such as hardness, toughness, flexibility, heat resistance, solubility, crystallization tendency, etc.

Coordination or Zeigler–Natta Polymerization



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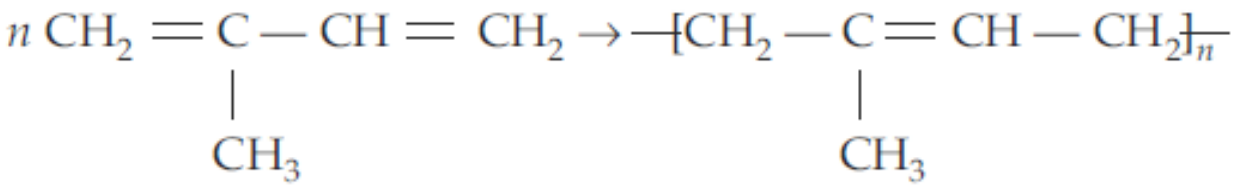
Preparation, properties and applications of synthetic rubbers:

- **Rubbers are non-crystalline in nature and are high polymers that have elastic properties.**
- **There are two types of rubber - natural rubber and synthetic rubber.**

Natural Rubber

- **The main source of natural rubber is the tree *Hevea brazilie*nsis.**
- **Rubber is obtained from latex collected from cuts made in the bark of the tree.**
- **During the treatment of latex, isoprene molecules in latex get polymerized to form poly isoprene.**



Isoprene

Poly-isoprene (*cis* form)

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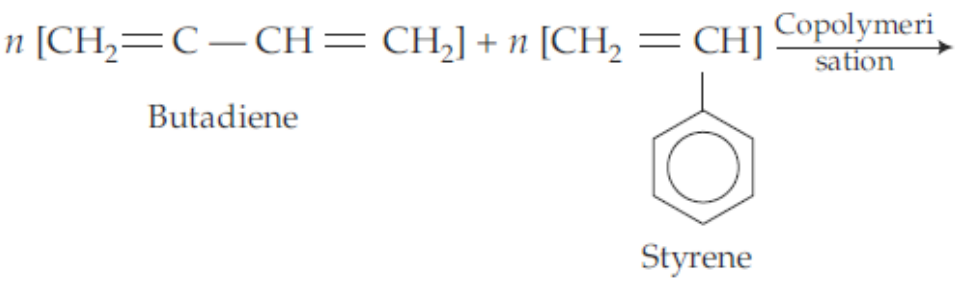
Synthetic Rubber

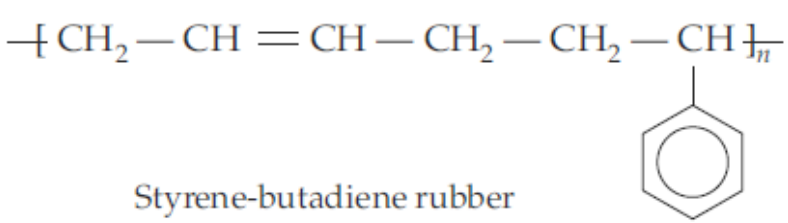
- The first synthetic rubber was originally developed in USA during the Second World War as a substitute for natural rubber.

- They are man-made rubbers which are superior to natural rubber in certain properties. Examples: Buna-S rubber, butyl neoprene rubber, nitrile rubber, etc.

Buna-S Rubber (SBR)

It is prepared by co-polymerization of 75% butadiene and 25% styrene at 0°C in the presence of a peroxide catalyst.





- **Properties:**
  - SBR on vulcanization gives a cross-linked polymer and possesses high resilience, high load-bearing capacity and high abrasion resistance.

- **Uses:**
  - It is highly useful in manufacture of motor tyres.
  - It is used in manufacture of floor tiles, cable insulations, etc.

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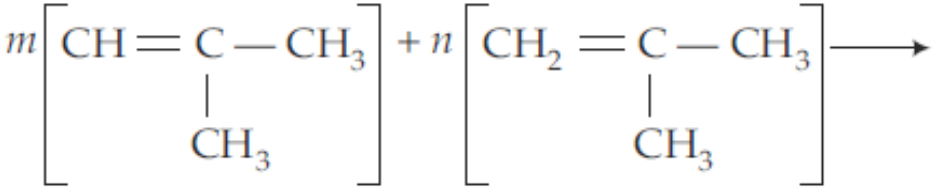
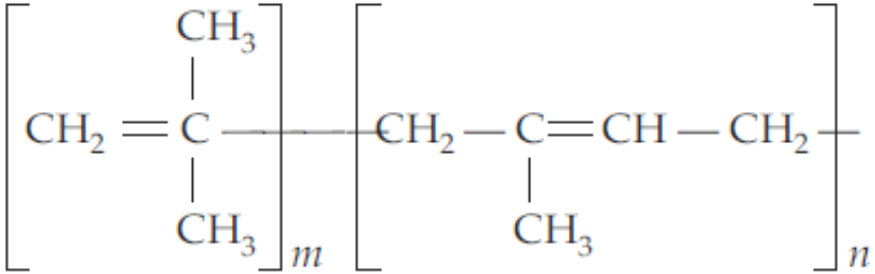
Synthetic Rubber

Butyl Rubber (GR-I)

Preparation: It is made by co-polymerization of isobutylenes with small amounts of isoprene.



Isobutene Isoprene



Polyisobutene-co-isoprene **(Butyl rubber)**

- Properties:
  - It is amorphous under normal condition but crystalline on stretching.
  - It is resistant to heat, chemicals, ozone, polar solvents.
  - It is soluble in nonpolar solvents such as benzene.

- Uses:
  - Butyl rubber is used in making cycle and automobile tubes.
  - It is used for making insulating cables.

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Silicones

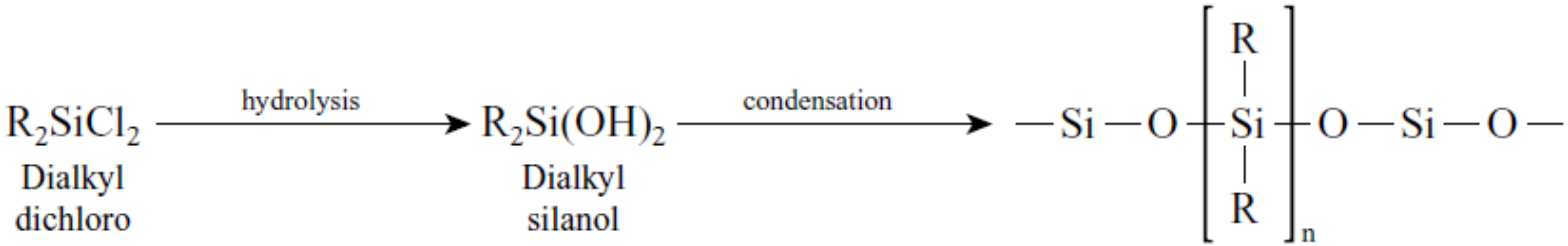
Silicones are synthetic polymers made from silicon and oxygen atoms. They are used in a variety of products, including sealants, adhesives, and lubricants.

Preparation:

- For preparation of silicones, dialkyl-substituted silanes are used as raw materials.
- They undergo hydrolysis and condensation polymerization to form silicone polymers.
- Silicones are of mainly two types: Straight-chain polymers and Cross-linked polymers.

Straight-chain polymers:

❖ Dialkyl-substituted silanes are subjected to hydrolysis and undergo condensation polymerization to form straight-chain thermoplastic silicones.



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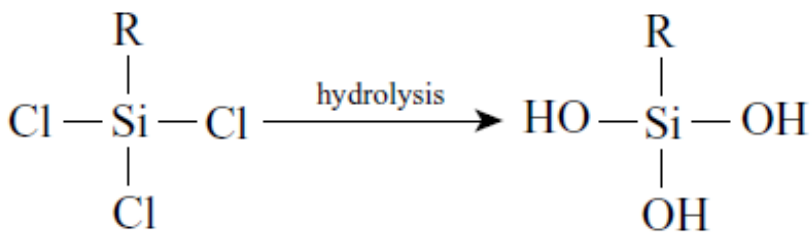
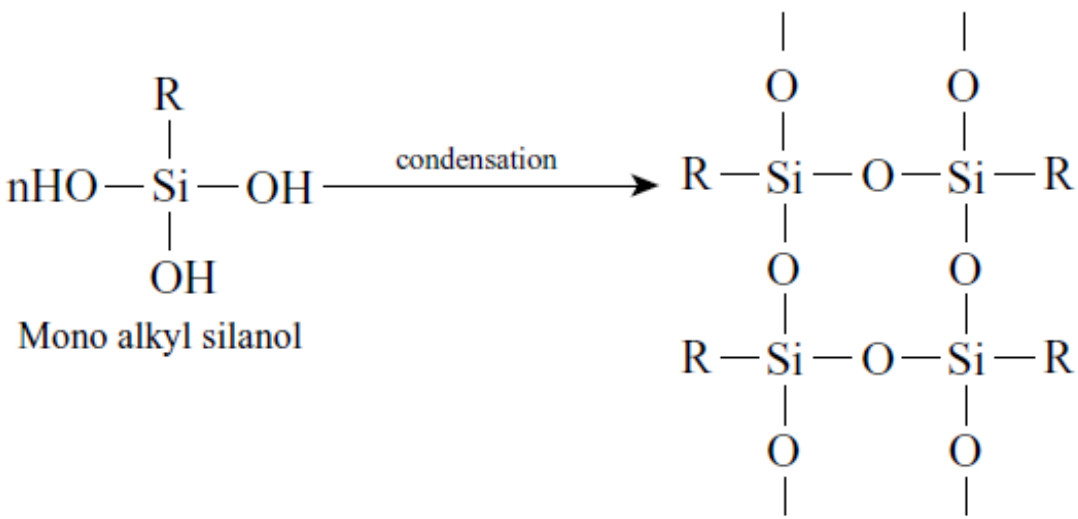
Silicones

Cross-linked polymers:

❖ Mono alkyl-substituted silanes are subjected to hydrolysis and proceeds condensation reaction gives cross-linked polymers.



Mono alkyl silanol



Mono alkyl silanol

Cross linked polymer

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Silicones

Properties

- Silicones have resistance to heat, cold, oxidation, etc.
- They obtain as liquids, resins and elastomers.
- Silicon liquids are having repulsion with water.
- They have also resistance with chemicals.
- Silicone rubber may form reaction between silicone resins, silica and alumina.

Uses

- Silicones are used as anti-forms in the preparation of food stuff.
- Used as elastics, insulators, and ceiling joints, and also used in medicine.
- They are used as surface-coating agents.
- Silicones are used for lamination.

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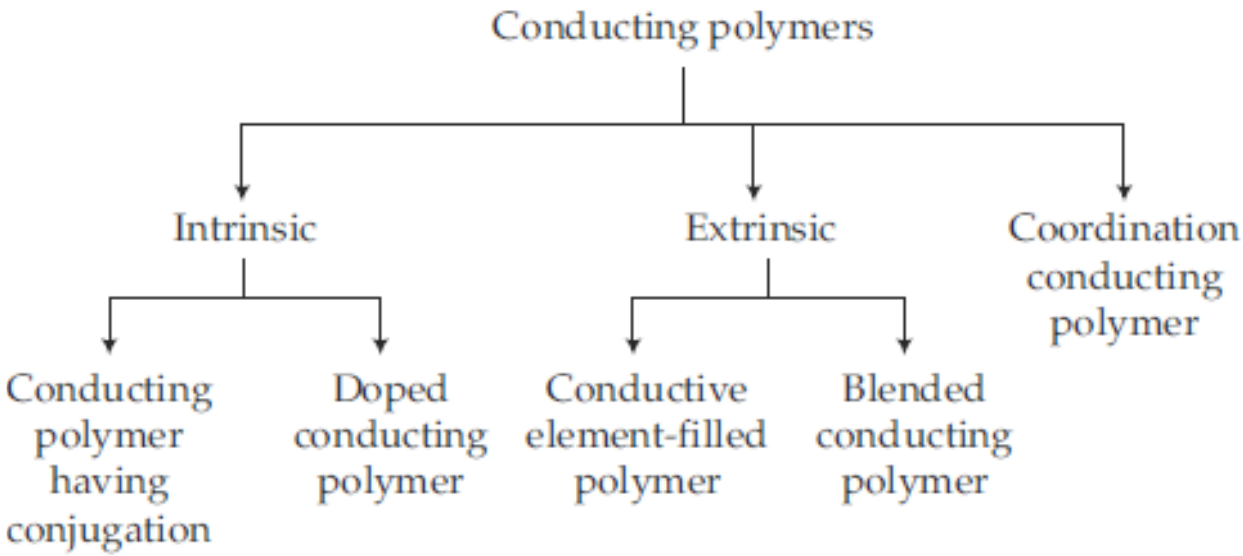
Conducting Polymers:

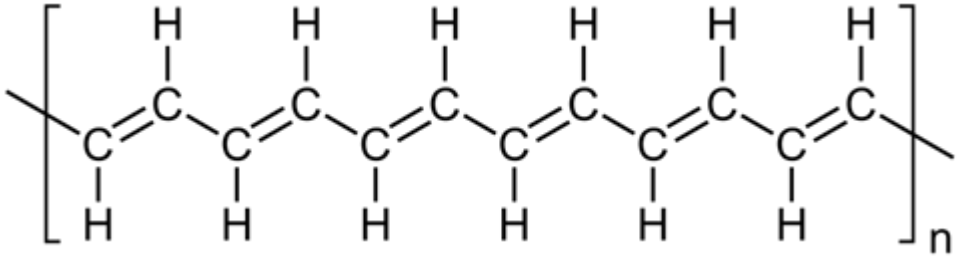
Conducting polymer is an organic polymer having highly delocalized π-electron system and electrical conductance.



Intrinsic Conducting Polymer (ICP): **These polymers have extensive conjugation in the backbone which is responsible for their conductance.**

Conducting Polymer having Conjugation: **Such a polymer contains a conjugated π-electron in the backbone which increases its conductivity to a large extent.**

Ex. Polyacetylene



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Conducting Polymers:

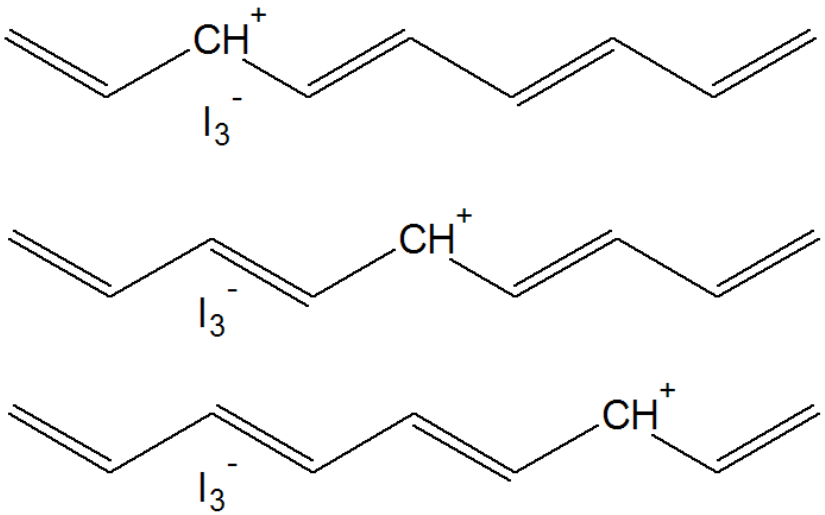
Intrinsic Conducting Polymer (ICP):

Doped Conducting Polym:

- Generally, intrinsic conducting polymers possess low conductivity.
- Conductivity can be increased by creating either positive or negative charge on the polymer backbone by oxidation or reduction. This is called doping and is of two types.

P-Doping:

- It involves treating the intrinsic conducting polymer with a Lewis acid.
- Oxidation takes place and a positive charge is created on the polymer backbone.
- Some of P-dopants used are $FeCl_3$, I_2, Br_2, etc.



Movement of radical cation ("polaron")

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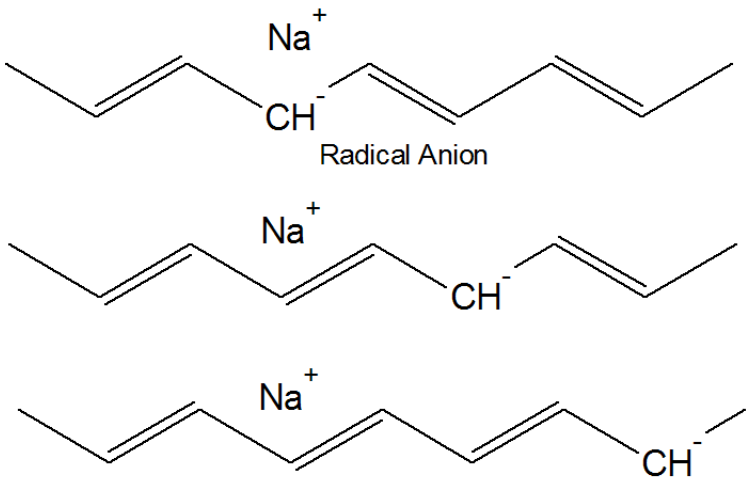
Conducting Polymers:

Intrinsic Conducting Polymer (ICP):

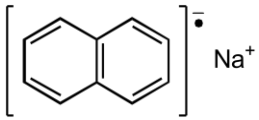
Doped Conducting Polym:

N-Doping:

- It involves treating the intrinsic conducting polymer with a Lewis base.
- Reduction takes place and a negative charge is created on the polymer backbone.
- Some of N-dopants used are sodium naphthalide.



Movement of radical anion ("polaron")



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Conducting Polymers:

Extrinsic Conducting Polymer (ECP):

In these polymers, conductivity is due to the presence of externally added substances.

Conductive Element-filled Polymer:

- In this, the polymer acts as a binder to hold the conducting element (such as carbon black, metallic fibres) together in solid entity. The minimum concentration of the conductive filler which should be added so that polymer starts conducting, is known as percolation threshold.

- These polymers possess good conductivity and are generally low in cost, light in weight, mechanically durable, strong and easily processable in different forms, shapes and sizes.

Blended Conducting Polymer:

- These polymers are obtained by blending a conventional polymer with a conducting polymer.

- Such polymers possess better physical, chemical, electrical and mechanical properties and can be easily processed.

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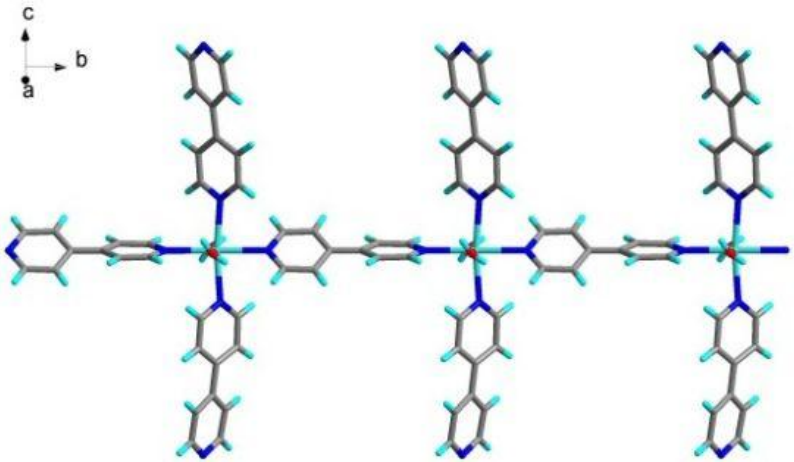
Conducting Polymers:

Coordination Conducting Polymers:

It is a charge transfer complex containing a polymer obtained by combining a metal atom with a polydentate ligand. Degree of polymerization in such polymers is small



Applications

- It is used in rechargeable light-weight batteries.
- It is used in wiring aircrafts and aerospace components.
- It is also used in solar panels.

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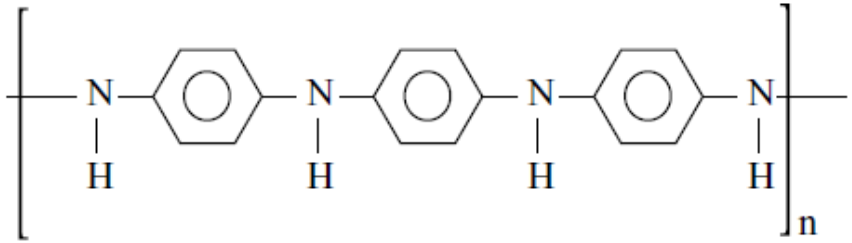
Conducting Polymers:

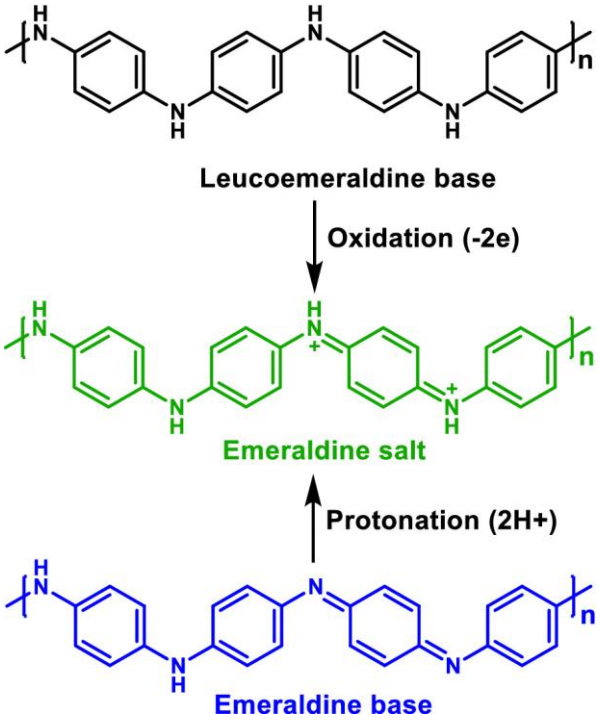


Electrically conducting polyaniline

Alan MacDiarmid

Properties:

- Transparent in thin layers and stable when heated
- Highly reactive, redox active material.
- Green in conducting state and changes its colour in different media.
- For example, under reducing condition, it becomes yellow and under oxidizing or basic condition, it becomes blue.



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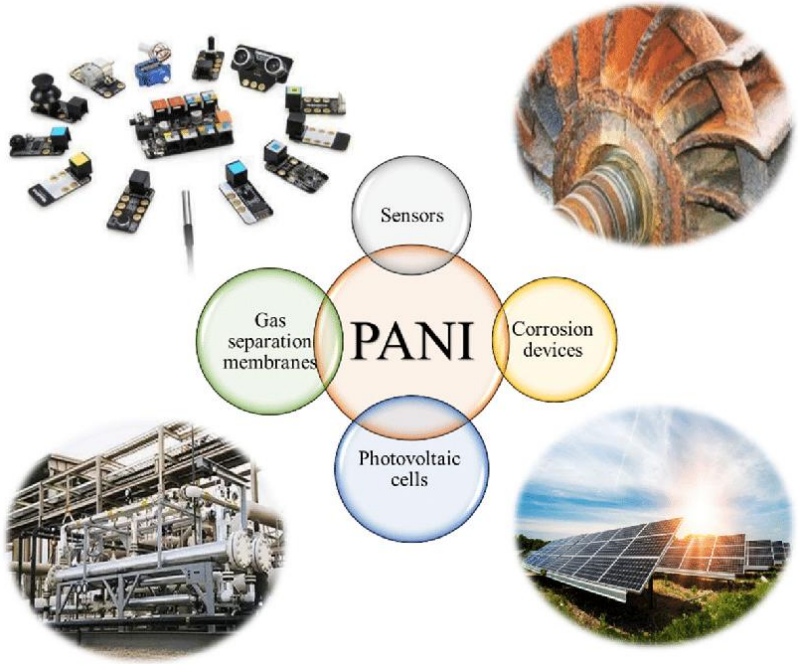
Conducting Polymers: Applications

- Due to its reversible electrochemical response during anodic oxidation and cathodic reduction, it is useful as a second electrode in rechargeable batteries and electrochromic display devices.

- It is used for coating films and semi-finished articles.

- It is also used for corrosion protection, sensors, smart windows, printed circuit boards, conductive pipes for explosives, and conductive fabrics.



Unit 4:Polymer Chemistry

Effect of Polymer Structure on Properties

The physical, chemical and mechanical properties of a polymer are determined by the nature of monomer units and its structure.

Strength

(i) Types of Attractive Forces: The individual chains in polymers are held by weak van der Waals forces of attraction. The strength can be increased by increasing the intermolecular van derWaals forces which can be greatly increased by the presence of the polar groups like –OH, -COOH, -Cl, -CN, etc.,

(ii) Molecular Weight: Polymers with low molecular weight tend to be soft or even fluidic with little mechanical strength, while those with high molecular weight are typically harder and more robust.

(iii) Slipping Power: Polymer molecules greatly affect the resistance to slip and consequent deformation of a polymer, and slipping power mainly depends on shape of the molecule.

Unit 4:Polymer Chemistry

Effect of Polymer Structure on Properties

Plastic deformation

On application of heat and pressure, there is a permanent deformation occurs on a polymer. This property is mainly useful to classify the polymers (thermoplastics or thermosetting plastics).

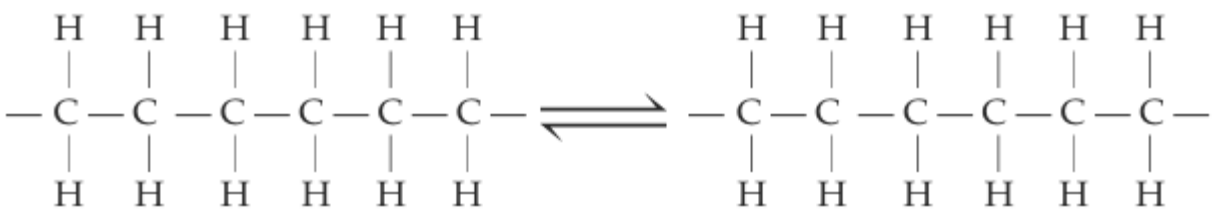
- Linear Polymer can be deformed by heat and pressure. This property of thermoplastics is called plastic deformation. Repeated heating and cooling does not affect the chemical nature of the polymer

- In giant cross-linked polymers, deformation does not occur on heating. Such types of polymers, once moulded, cannot be remoulded and they are called thermosetting plastics. Therefore, in thermosetting polymers, plasticity does not increase with increase in temperature.

(a) Plastic Deformation in a Linear Polymer

These are polymeric substances which can be deformed by heat and pressure. This property of thermoplastics is called ***plastic deformation***. Repeated heating and cooling does not affect the chemical nature of the polymer.

Linear polymers show greatest degree of plastic deformation, as the molecules are held by weak van derWaals forces of attraction. Under high temperature and pressure, the forces become weaker. So slippage occurs between the chains. On cooling, they regain their original hardness. Repeated heating and cooling do not affect the nature of the polymer. So plasticity of such polymers is **reversible** and can be remoulded many times. They are called ***thermoplastics***, e.g. **Polyethylene.**



(b) Plastic Deformation in Cross-linked Polymer

In giant cross-linked polymers, deformation does not occur on heating because only primary covalent bonds are present throughout the entire structure. Due to the presence of strong covalent bonds, it cannot be easily affected either by heat or pressure. So no slippage occurs in between the chains. Such types of polymers, once moulded, cannot be remoulded and they are called ***thermosetting plastics***. **Therefore, in thermosetting polymers, plasticity does not increase with increase in temperature.**

Unit 4:Polymer Chemistry

Effect of Polymer Structure on Properties

Crystallinity

(a) Amorphous Polymer: They have random arrangement of polymer chains and hence the intermolecular forces are weak. Polymers with long chains and bulky side groups of low degree of symmetry, do not crystallise easily and, therefore, form an amorphous state.

(b) Crystalline Polymer: Crystallization imparts denser packing of molecules, thereby increasing the intermolecular forces of attraction. The polymer chains are regularly arranged in a crystalline polymer. So the intermolecular forces are increased, resulting in high density, greater strength and sharper softening point.

(c) Crystallites Embedded in an Amorphous Matrix: Most of the polymers have crystallites embedded in an amorphous matrix. In such cases, crystallites provide required hardness, rigidity and heat resistance while the amorphous matrix provides flexibility. Co-polymerization decreases crystallinity. The relative extent of crystallites and amorphous region can be adjusted to get the required hardness and flexibility.

Unit 4:Polymer Chemistry

Effect of Polymer Structure on Properties

Chemical Resistance

The chemical resistance of a polymer is decided by the nature of monomeric units and their molecular arrangement. Chemical attack causes softening, swelling and loss of strength of material

Other factors such as molecular weight, crystallinity and degree of cross linking also determine the chemical resistance.

(i) The tendency to be soluble in a particular solvent decreases with increase of molecular weight of the polymer.

(ii) The chemical resistance of crystalline polymers is more than that of amorphous polymers.

(iii) Greater the degree of cross linking in the polymer, lesser is its solubility.

Unit 4:Polymer Chemistry

Polymer Blends And Alloys

Polymer Blends

A polymer blend is a simple physical combination of two or more incompatible polymers, e.g. wood is a natural blend of more than one polymer.

Properties

- A polymer blend appears as separate phases, when viewed under a microscope.
- Between different polymeric chains, only van der Waals forces, dipole interaction or hydrogen bonding exists.
- Blending usually improves the properties like processability, impact strength, abrasion resistance, flame retardance, etc.
- Sensitive component of a blend may be protected from degradation by blending. PMMA and SAN

Applications

- Polycarbonate (PC)-ABS (acrylonitrile-butadiene-styrene) blend is used for making electrical housings and machine parts.
- Nylon 6-PC blend possesses improved thoughness and is employed for making sports equipment and transport containers.

Unit 4:Polymer Chemistry

Polymer Blends And Alloys

Polymer Alloy

Polymer alloy is a compatible mixture of two or more polymers, which interact chemically in the presence of compatibilizers under specific conditions of composition, temperature and pressure.

Properties

- Alloying of two polymers is, usually, done for achieving a specific combination of properties which are not attainable by the use of either of constituent polymers.
- Alloying of polymers, usually, improves properties like processability, abrasion resistance, impact strength, flame retardance, etc.

Examples

ABS-PC Alloys

ABS-PC alloys possess high softening temperature (110–130°C), good impact strength, good flexural strength, flame retardance and can be electroplated.

Uses: For making typewriters, electronic printers, wheel caps, food mixers, helmets, grills, pump casting and in the medical fi eld as a substitute for stainless steel, since it can be sterilized.